University of Jeddah
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CAR INSURANCE CLAIM PREDICTION

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Milestone 1

Problem Statement:

Car insurance companies face significant financial risks due to the uncertainty of claim occurrences. Identifying customers who are more likely to file a claim allows insurers to improve risk assessment, set fair premiums, and reduce potential losses. This project aims to analyze customer demographics, driving history, and vehicle information to build a predictive model that classifies whether a policyholder will file a claim or not.

Objectives:

1. Identify key factors influencing the likelihood of filing an insurance claim.
2. Analyze customer attributes such as age, income, credit score, vehicle type, and driving experience.
3. Build and evaluate predictive models (Decision Tree, Naïve Bayes, Logistic Regression) to classify claim outcomes.
4. Provide insights that help insurance companies make data-driven decisions to minimize risk.

Hypotheses:

H1: Younger and less experienced drivers have a higher probability of filing insurance claims
H2: Higher credit scores are associated with a lower likelihood of claims
H3: Drivers with a history of speeding or past accidents are more likely to file a claim
H4: Certain vehicle types (e.g., sports cars) are more prone to claims than sedans

Project Questions

1. What demographic and behavioral factors influence the probability of filing an insurance claim?
2. Which machine learning algorithm provides the most accurate prediction?
3. How do credit score, annual mileage, and vehicle type impact claim probability?
4. Can predictive modeling improve the fairness and accuracy of insurance policy decisions?

Data Source:

The dataset used in this project is titled “Car Insurance Claims”, collected from Kaggle. It contains both categorical and numerical variables related to policyholders, including:

- Demographic Information: Age, Gender, Education, Income.
- Driving Information: Driving Experience, Speeding Violations, Past Accidents.
- Vehicle Information: Vehicle Type, Vehicle Age, Annual Mileage, Vehicle Ownership.
- Financial Indicators: Credit Score.
- Target Variable: Outcome (0 = No Claim, 1 = Claim).
- The dataset contains more than 300 records and around 13 features, making it suitable for classification and clustering tasks in SAS Enterprise Miner

Car_Insurance_Claims_Dataset.csv:

https://drive.google.com/file/d/12FXrMmcOpS504Wl3SIfYBwKfq-Fy7qev/view?usp=sharing&utm_source=.com

Justification for Dataset Selection:

This dataset meets all project requirements:

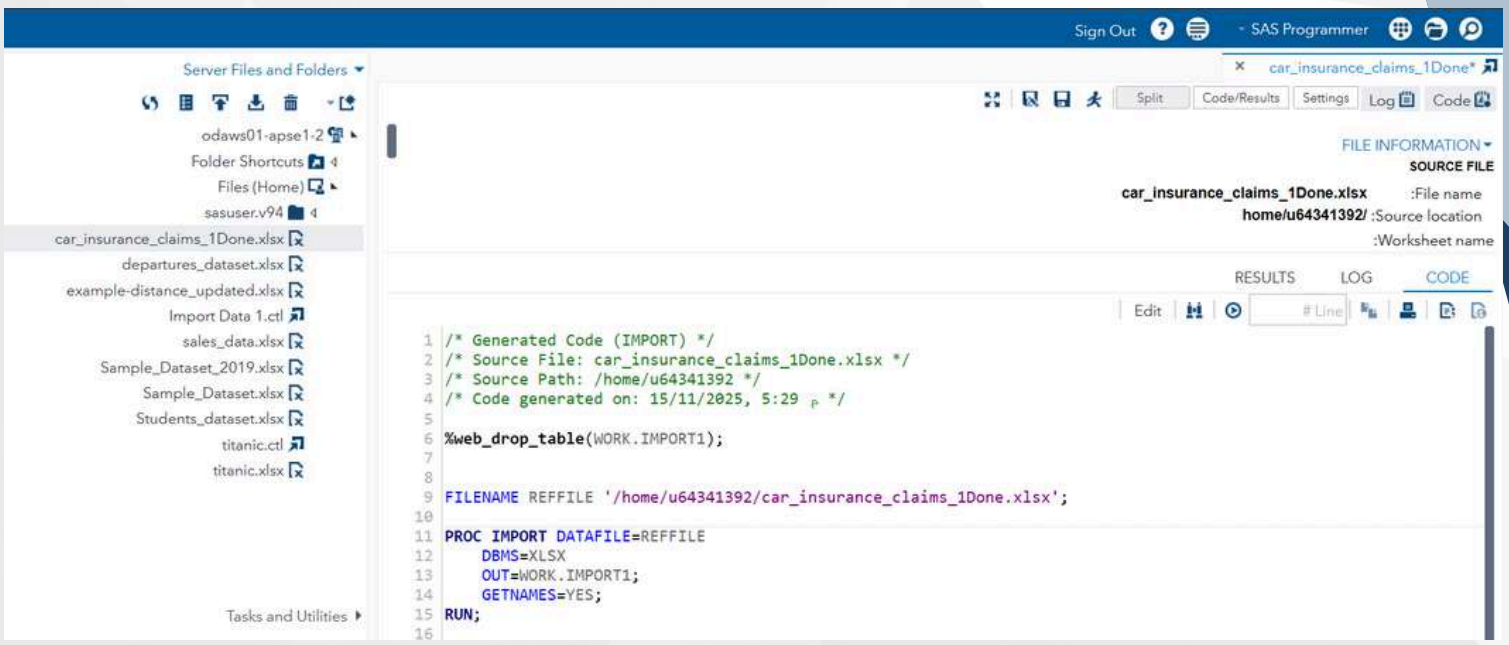
- It includes a mix of numerical and categorical variables.
- The target variable (OUTCOME) enables classification modeling.
- The data volume is sufficient for training and testing.
- It allows performing all project phases: data preparation, exploratory analysis, decision tree modeling, Naïve Bayes classification, and K-Means clustering.
- The topic is realistic, relevant, and aligned with data mining objectives in business and risk management.

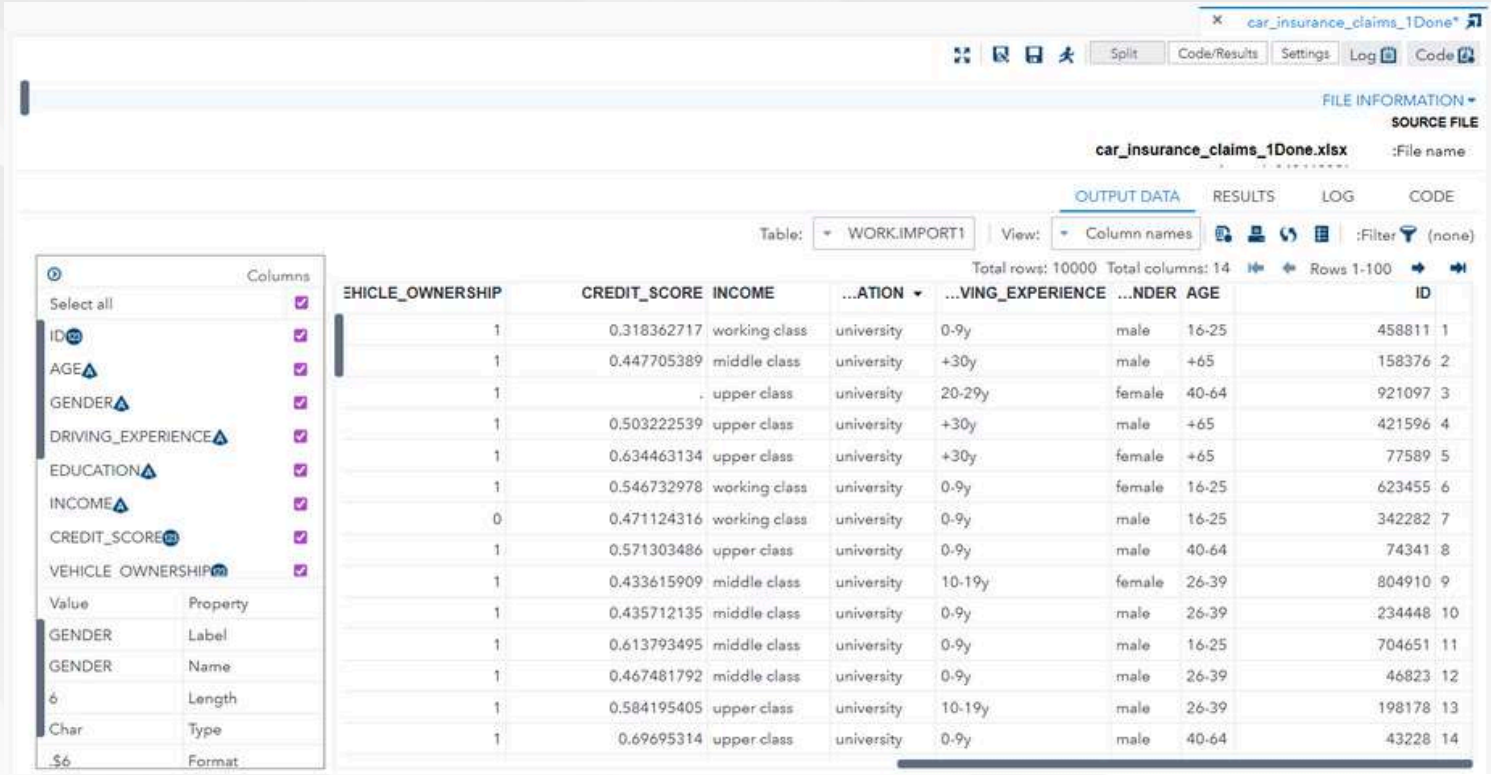
Conclusion

The proposed project will apply data mining techniques to predict car ,insurance claims using SAS Enterprise Miner. Through data cleaning exploratory analysis, and model building, the team will identify significant -risk factors and demonstrate how predictive analytics can enhance decision .making in the insurance industry

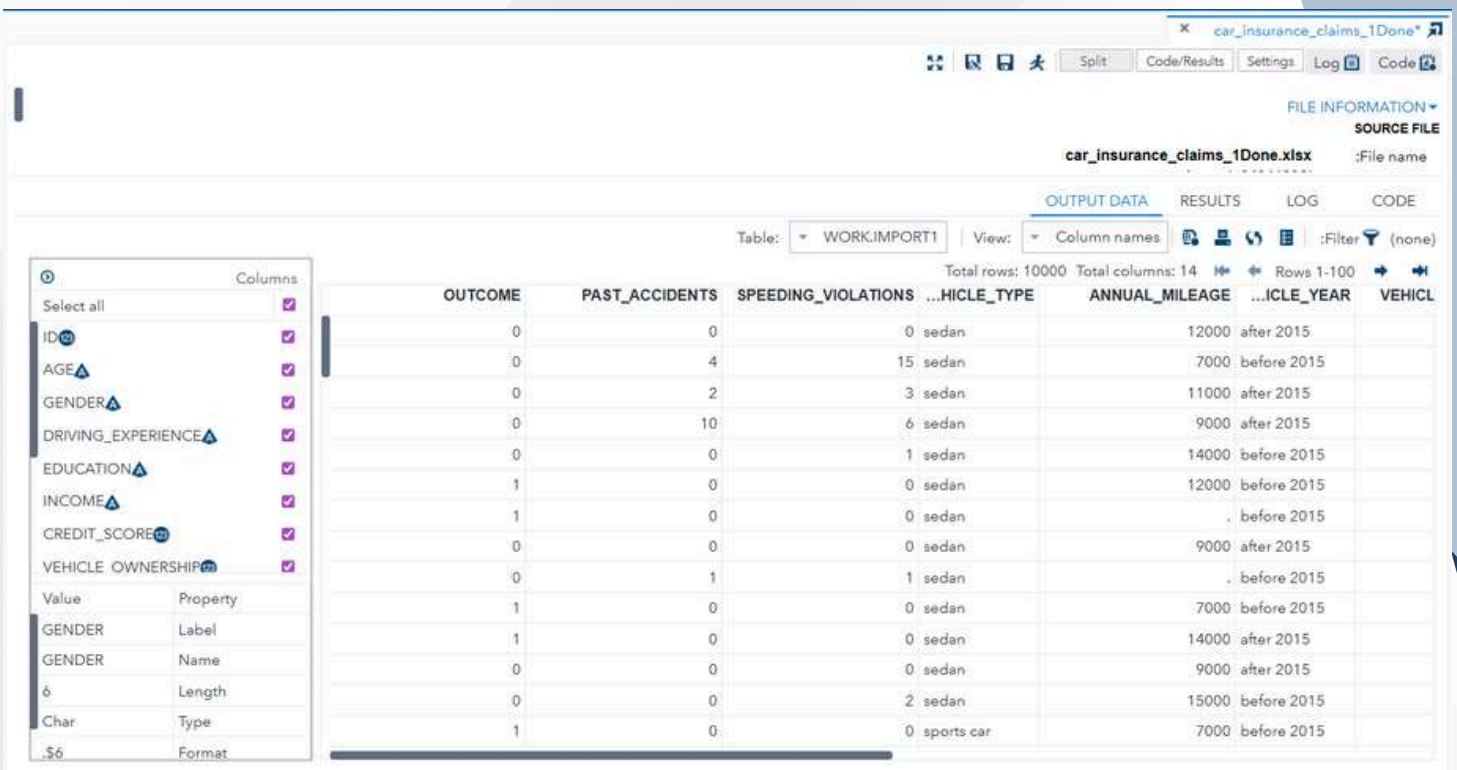
Milestone 2

Import the Dataset:



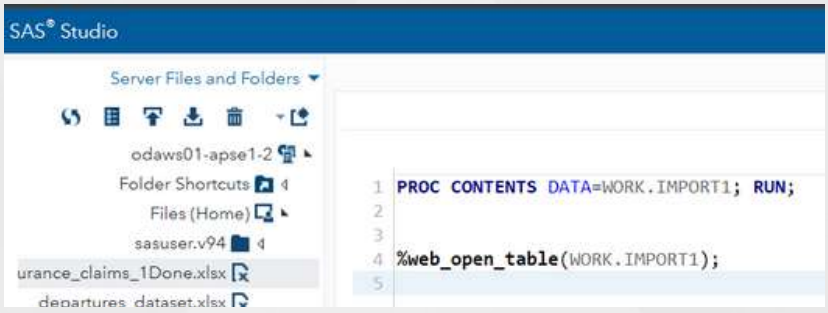


	EHICLE_OWNERSHIP	CREDIT_SCORE	INCOME	...ATION	...VING_EXPERIENCE	...NDER	AGE	ID	
	1	0.318362717	working class	university	0-9y	male	16-25	458811	1
	1	0.447705389	middle class	university	+30y	male	+65	158376	2
	1	.	upper class	university	20-29y	female	40-64	921097	3
	1	0.503222539	upper class	university	+30y	male	+65	421596	4
	1	0.634463134	upper class	university	+30y	female	+65	77589	5
	1	0.546732978	working class	university	0-9y	female	16-25	623455	6
	0	0.471124316	working class	university	0-9y	male	16-25	342282	7
	1	0.571303486	upper class	university	0-9y	male	40-64	74341	8
	1	0.433615909	middle class	university	10-19y	female	26-39	804910	9
	1	0.435712135	middle class	university	0-9y	male	26-39	234448	10
	1	0.613793495	middle class	university	0-9y	male	16-25	704651	11
	1	0.467481792	middle class	university	0-9y	male	26-39	46823	12
	1	0.584195405	upper class	university	10-19y	male	26-39	198178	13
	1	0.69695314	upper class	university	0-9y	male	40-64	43228	14



OUTCOME	PAST_ACCIDENTS	SPEEDING_VIOLATIONS	...HICLE_TYPE	ANNUAL_MILEAGE	...ICLE_YEAR	VEHICL
0	0	0	sedan	12000	after 2015	
0	4	15	sedan	7000	before 2015	
0	2	3	sedan	11000	after 2015	
0	10	6	sedan	9000	after 2015	
0	0	1	sedan	14000	before 2015	
1	0	0	sedan	12000	before 2015	
1	0	0	sedan	.	before 2015	
0	0	0	sedan	9000	after 2015	
0	1	1	sedan	.	before 2015	
1	0	0	sedan	7000	before 2015	
1	0	0	sedan	14000	after 2015	
0	0	0	sedan	9000	after 2015	
0	0	2	sedan	15000	before 2015	
1	0	0	sports car	7000	before 2015	

Identify Data Types :



Alphabetic List of Variables and Attributes

Label	Informat	Format	Len	Type	Variable	#
AGE	$5	$5	5	Char	AGE	2
ANNUAL_MILEAGE		BEST	8	Num	ANNUAL_MILEAGE	10
CREDIT_SCORE		BEST	8	Num	CREDIT_SCORE	7
DRIVING_EXPERIENCE	$6	$6	6	Char	DRIVING_EXPERIENCE	4
EDUCATION	$11	$11	11	Char	EDUCATION	5
GENDER	$6	$6	6	Char	GENDER	3
ID		BEST	8	Num	ID	1
INCOME	$13	$13	13	Char	INCOME	6
OUTCOME		BEST	8	Num	OUTCOME	14
PAST_ACCIDENTS		BEST	8	Num	PAST_ACCIDENTS	13
SPEEDING_VIOLATIONS		BEST	8	Num	SPEEDING_VIOLATIONS	12
VEHICLE_OWNERSHIP		BEST	8	Num	VEHICLE_OWNERSHIP	8
VEHICLE_TYPE	$10	$10	10	Char	VEHICLE_TYPE	11
VEHICLE_YEAR	$11	$11	11	Char	VEHICLE_YEAR	9

Explore the Data:

Distribution of Values for Categorical Variables before cleaning:

Using PROC FREQ, we generated frequency distributions for all categorical variables.
This output helped us identify repeated values, rare categories, and overall balance in each variable.
The table shows the frequency and percentage distribution for the ID variable before cleaning.

```
PROC FREQ DATA=WORK.IMPORT1;
TABLES _ALL_;
RUN;
```

The FREQ Procedure

ID				
Cumulative Percent	Cumulative Frequency	Percent	Frequency	ID
0.01	1	0.01	1	101
0.02	2	0.01	1	125
0.03	3	0.01	1	166
0.04	4	0.01	1	186
0.05	5	0.01	1	217
0.06	6	0.01	1	226
0.07	7	0.01	1	244
0.08	8	0.01	1	286
0.09	9	0.01	1	381
0.10	10	0.01	1	542
0.11	11	0.01	1	603
0.12	12	0.01	1	775
0.13	13	0.01	1	793
0.14	14	0.01	1	864

AGE				
Cumulative Percent	Cumulative Frequency	Percent	Frequency	AGE
20.16	2016	20.16	2016	16-25
50.79	5079	30.63	3063	26-39
80.10	8010	29.31	2931	40-64
100.00	10000	19.90	1990	+65

GENDER				
Cumulative Percent	Cumulative Frequency	Percent	Frequency	GENDER
50.10	5010	50.10	5010	female
100.00	10000	49.90	4990	male

DRIVING_EXPERIENCE				
Cumulative Percent	Cumulative Frequency	Percent	Frequency	DRIVING_EXPERIENCE
35.30	3530	35.30	3530	0-9y
68.29	6829	32.99	3299	10-19y
89.48	8948	21.19	2119	20-29y
100.00	10000	10.52	1052	+30y

EDUCATION				
Cumulative Percent	Cumulative Frequency	Percent	Frequency	EDUCATION
41.57	4157	41.57	4157	high school
60.72	6072	19.15	1915	none
100.00	10000	39.28	3928	university

INCOME				
Cumulative Percent	Cumulative Frequency	Percent	Frequency	INCOME
21.38	2138	21.38	2138	middle class
39.52	3952	18.14	1814	poverty
82.88	8288	43.36	4336	upper class
100.00	10000	17.12	1712	working class

CREDIT_SCORE				
Cumulative Percent	Cumulative Frequency	Percent	Frequency	CREDIT_SCORE
0.01	1	0.01	1	0.053357545
0.02	2	0.01	1	0.060866616
0.03	3	0.01	1	0.064810353
0.04	4	0.01	1	0.095387094
0.06	5	0.01	1	0.097211
0.07	6	0.01	1	0.099655226
0.08	7	0.01	1	0.112631117
0.09	8	0.01	1	0.115402476
0.10	9	0.01	1	0.116263801
0.11	10	0.01	1	0.118445443
0.12	11	0.01	1	0.119144798
0.13	12	0.01	1	0.121826084
0.14	13	0.01	1	0.123794616
0.16	14	0.01	1	0.123977549
0.17	15	0.01	1	0.125876021
0.18	16	0.01	1	0.125895544
0.19	17	0.01	1	0.126453431

99.94	9013	0.01	1	0.874829991
99.96	9014	0.01	1	0.884409527
99.97	9015	0.01	1	0.89629944
99.98	9016	0.01	1	0.914116859
99.99	9017	0.01	1	0.915883183
100.00	9018	0.01	1	0.960818828
Frequency Missing = 982				

VEHICLE_OWNERSHIP				
Cumulative Percent	Cumulative Frequency	Percent	Frequency	VEHICLE_OWNERSHIP
30.30	3030	30.30	3030	0
100.00	10000	69.70	6970	1

VEHICLE_YEAR				
Cumulative Percent	Cumulative Frequency	Percent	Frequency	VEHICLE_YEAR
30.33	3033	30.33	3033	after 2015
100.00	10000	69.67	6967	before 2015

ANNUAL_MILEAGE				
Cumulative Percent	Cumulative Frequency	Percent	Frequency	ANNUAL_MILEAGE
0.02	2	0.02	2	2000
0.13	12	0.11	10	3000
0.43	39	0.30	27	4000
1.15	104	0.72	65	5000
2.97	269	1.82	165	6000
6.54	591	3.56	322	7000
12.69	1148	6.16	557	8000
22.16	2004	9.47	856	9000
34.00	3075	11.84	1071	10000
47.86	4328	13.86	1253	11000
61.33	5546	13.47	1218	12000
73.90	6683	12.57	1137	13000
83.79	7577	9.89	894	14000
90.78	8209	6.99	632	15000
95.41	8628	4.63	419	16000
98.13	8874	2.72	246	17000
99.27	8977	1.14	103	18000
99.80	9025	0.53	48	19000
99.94	9038	0.14	13	20000
99.98	9041	0.03	3	21000
100.00	9043	0.02	2	22000
Frequency Missing = 957				

VEHICLE_TYPE				
Cumulative Percent	Cumulative Frequency	Percent	Frequency	VEHICLE_TYPE
95.23	9523	95.23	9523	sedan
100.00	10000	4.77	477	sports car

SPEEDING_VIOLATIONS				
Cumulative Percent	Cumulative Frequency	Percent	Frequency	SPEEDING_VIOLATIONS
50.28	5028	50.28	5028	0
65.72	6572	15.44	1544	1
77.33	7733	11.61	1161	2
85.63	8563	8.30	830	3
90.93	9093	5.30	530	4
94.12	9412	3.19	319	5
96.00	9600	1.88	188	6
97.40	9740	1.40	140	7
98.15	9815	0.75	75	8
98.64	9864	0.49	49	9
99.14	9914	0.50	50	10

PAST_ACCIDENTS				
Cumulative Percent	Cumulative Frequency	Percent	Frequency	PAST_ACCIDENTS
55.84	5584	55.84	5584	0
73.67	7367	17.83	1783	1
84.71	8471	11.04	1104	2
91.17	9117	6.46	646	3
94.83	9483	3.66	366	4
97.15	9715	2.32	232	5
98.59	9859	1.44	144	6
99.20	9920	0.61	61	7
99.61	9961	0.41	41	8
99.80	9980	0.19	19	9
99.89	9989	0.09	9	10
99.96	9996	0.07	7	11
99.98	9998	0.02	2	12
99.99	9999	0.01	1	14
100.00	10000	0.01	1	15

OUTCOME				
Cumulative Percent	Cumulative Frequency	Percent	Frequency	OUTCOME
68.67	6867	68.67	6867	0
100.00	10000	31.33	3133	1

Distribution of Values for Numeric Variables before cleaning:

```
PROC MEANS DATA=WORK.IMPORT1 N MEAN MEDIAN STD MIN MAX;
VAR _NUMERIC_;
RUN;
```

The MEANS Procedure

Maximum	Minimum	Std Dev	Median	Mean	N	Label	Variable
999976.00	101.0000000	290030.77	501777.00	500521.91	10000	ID	ID
0.9608188	0.0533575	0.1376881	0.5250328	0.5158128	9018	CREDIT_SCORE	CREDIT_SCORE
22000.00	2000.00	2818.43	12000.00	11697.00	9043	ANNUAL_MILEAGE	ANNUAL_MILEAGE
22.0000000	0	2.2419657	0	1.4829000	10000	SPEEDING_VIOLATIONS	SPEEDING_VIOLATIONS
15.0000000	0	1.6524537	0	1.0563000	10000	PAST_ACCIDENTS	PAST_ACCIDENTS

Using PROC MEANS, we generated summary statistics for all numeric variables before cleaning to understand their initial distribution.
,The output revealed missing values, unusual minimum/maximum values and inconsistencies that indicate the need for data cleaning
This step helped identify which variables require type correction and preparation before further analysis

Data Cleaning :

Converting Variables to the Correct Data Types :

```
data WORK.IMPORT1;
    set IMPORT1;

    /* Convert ID to numeric */
    ID_num = input(ID, 12.);
    /* Convert ANNUAL_MILEAGE to numeric (some values are empty) */
    Mileage = input(ANNUAL_MILEAGE, best12.);

       /*Convert numeric fields directly */
    Credit = CREDIT_SCORE;
    Speed_Viol = SPEEDING_VIOLATIONS;
    Accidents = PAST_ACCIDENTS;

    /* Binary variables */
    Veh_Own = VEHICLE_OWNERSHIP;
    Outcome_bin = OUTCOME;

    /* Categorical variables: keep as character */
    Age_cat     = strip(AGE);
    Gender      = strip(GENDER);
    Drive_exp   = strip(DRIVING_EXPERIENCE);
    Education   = strip(EDUCATION);
    Income_cat  = strip(INCOME);
    Veh_year    = strip(VEHICLE_YEAR);
    Veh_type    = strip(VEHICLE_TYPE);

    /* Keep clean variables only */
    keep ID_num Age_cat Gender Drive_exp Education Income_cat Credit Veh_Own
         Veh_year Mileage Veh_type Speed_Viol Accidents Outcome_bin;
run;
```

Table: WORK.IMPORT1 View: Column names :Filter (none)

Total rows: 10000 Total columns: 14 Rows 1-100

:idents	Speed_Viol	Credit	Mileage	ID_num	EDUCATION	...NDER	
3	6	0.40938717	14000	647692	none	male	1
4	15	0.447705389	7000	158376	university	male	2
0	0	0.654721892	.	581821	university	female	3
2	1	0.445405569	14000	407928	high school	male	4
3	0	0.509224941	13000	562979	high school	female	5
5	4	0.565246547	8000	595172	university	female	6
0	0	0.576121835	13000	16453	none	female	7
0	0	0.534287706	13000	325516	none	female	8
2	1	0.746672171	17000	68125	university	male	9
2	4	0.618378258	11000	471505	high school	female	10
0	0	0.77478659	11000	653167	university	female	11
2	3	0.583202776	9000	555993	university	female	12
3	14	0.591927307	9000	424145	university	male	13
3	1	0.631807266	11000	179805	university	female	14
1	2	0.685876328	14000	809541	university	female	15
0	0	0.641599291	13000	859256	university	female	16
1	0	.	9000	309935	high school	female	17
3	2	0.491675067	11000	902327	high school	male	18
0	0	0.562364576	9000	110873	high school	male	19

Value	Property
Credit	Label
Credit	Name
8	Length
Numeric	Type
	Format
	Informat

Imputing Missing Values Using the Mean :

```
proc stdize data=WORK.IMPORT1 out=WORK.IMPORT1 reponly;
    var Credit Mileage;
run;
```

Program 1 | car_insurance_claims_1Done*

OUTPUT DATA | RESULTS | LOG | CODE

Table: WORK.IMPORT1 | View: Column names | :Filter (none)

Total rows: 10000 Total columns: 14 | Rows 1-100

Columns

Select all

GENDER
EDUCATION
ID_num
Mileage
Credit
Speed_Viol
Accidents
Veh_Own
Outcome_bin
Age_cat
Drive_exp

Value	Property
	Label
	Name
	Length
	Type
	Format
	Informat

:idents	Speed_Viol	Credit	Mileage	ID_num	EDUCATION	...NDER	
0	1	0.676148253	13000	7905	university	female	79
3	4	0.324743831	10000	7967	high school	male	80
2	4	0.567526303	15000	8023	university	male	81
0	0	0.685929636	13000	8055	none	female	82
0	0	0.519322805	11697.003207	8309	high school	male	83
2	1	0.720087446	12000	8316	high school	male	84
1	2	0.425372617	9000	8388	none	female	85
2	3	0.5158128096	11697.003207	8464	university	male	86
0	0	0.248401215	10000	8488	university	female	87
1	0	0.503790021	12000	8539	high school	male	88
0	0	0.500158714	9000	8605	university	female	89
0	1	0.70487148	8000	8618	university	female	90
0	3	0.668981211	11697.003207	8852	high school	female	91
4	2	0.564567537	12000	8982	university	male	92
2	3	0.625146635	17000	9013	high school	male	93
1	1	0.474348001	9000	9128	high school	female	94
0	0	0.745420167	13000	9204	university	female	95
1	3	0.61366119	9000	9251	high school	female	96
0	0	0.633817535	7000	9289	university	female	97

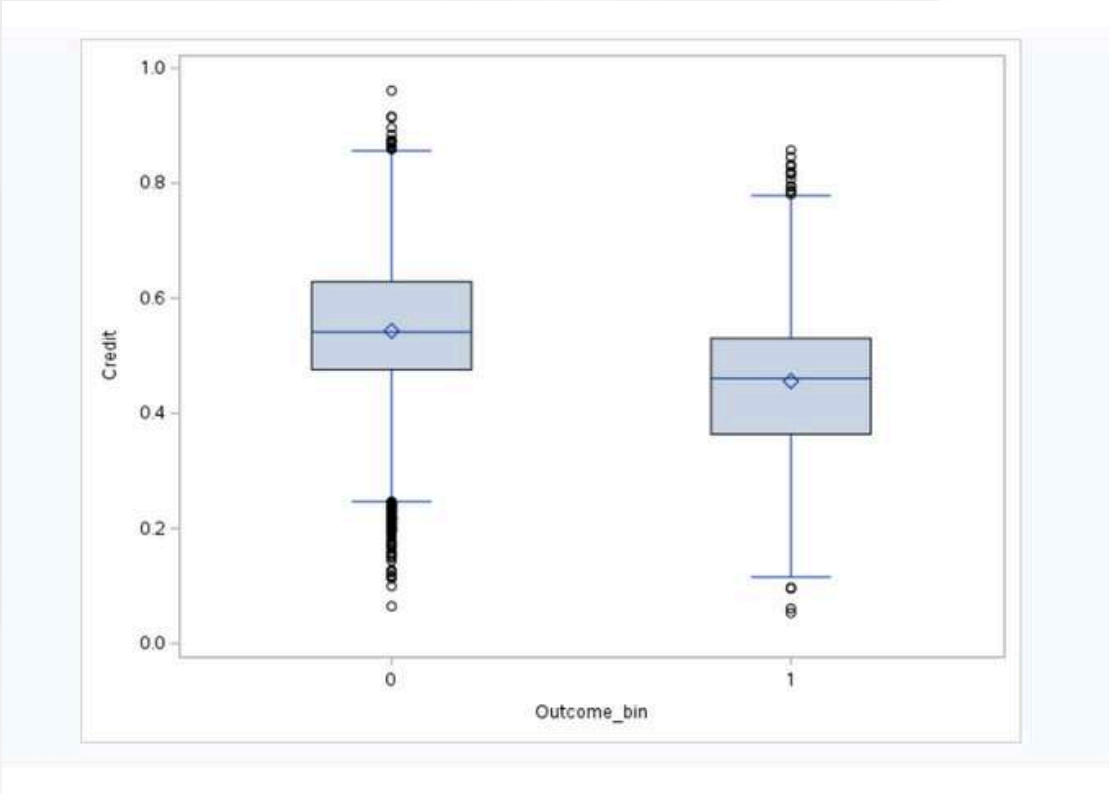
We used the PROC STDIZE procedure to replace missing values in the numeric variables Credit and .Mileage with their respective mean values

This method ensures that missing data does not reduce the number of observations and helps .maintain the dataset's consistency

The updated output table confirms that previously missing entries have been successfully imputed with .mean-based replacement

Detecting Outliers Using BoxpLot :

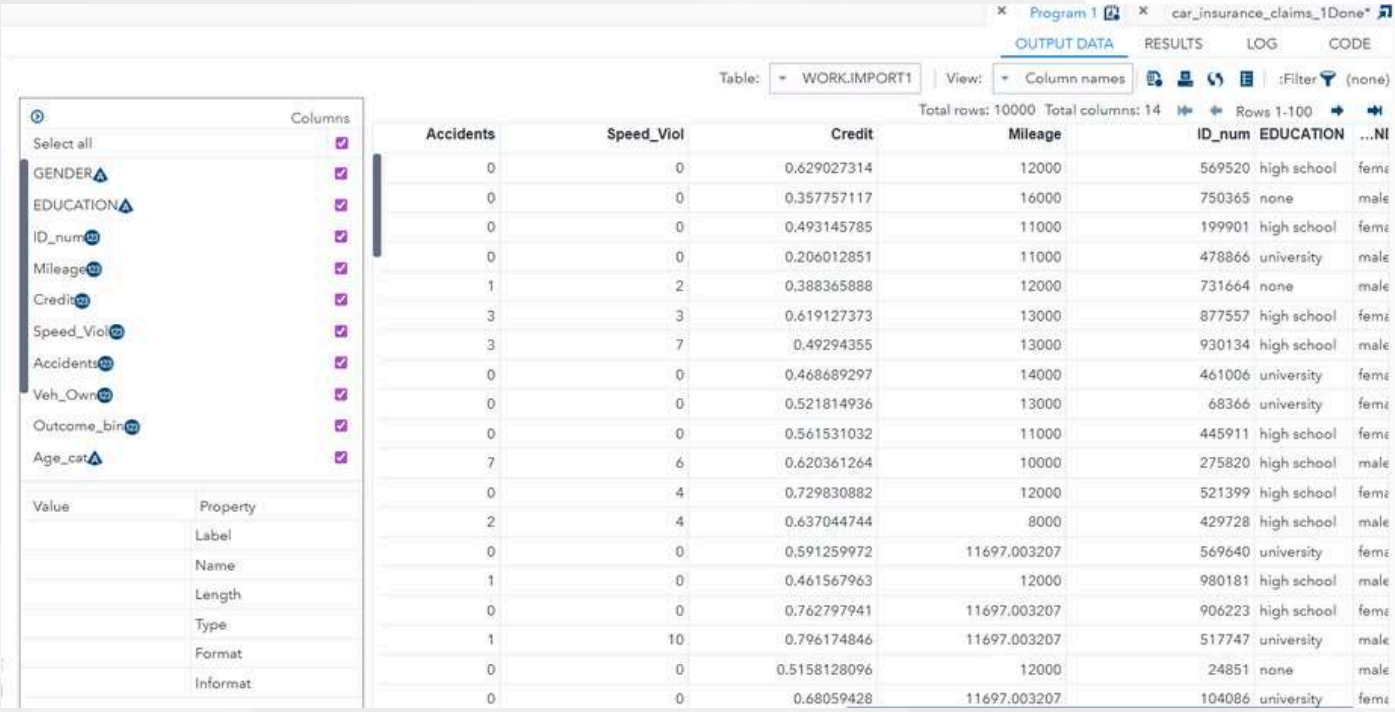
```
proc sgplot data=WORK.IMPORT1;
    vbox Credit / category=Outcome_bin;
run;
```



We chose the variables Credit and Outcome_bin because Credit is one of the most important variables indicating risk in insurance, and Outcome_bin is the variable that determines whether a claim was made or not. The box plot compares the distribution of credit scores between people who filed a claim and those who did not, helping to analyze the relationship between them.

Handling Outliers :

```
data WORK.IMPORT1;
    set WORK.IMPORT1;
    if Credit > 0.9 then Credit = 0.9;
    if Credit < 0.1 then Credit = 0.1;
run;
```



Accidents	Speed_Viol	Credit	Mileage	ID_num	EDUCATION	...NI
0	0	0.629027314	12000	569520	high school	fema
0	0	0.357757117	16000	750365	none	male
0	0	0.493145785	11000	199901	high school	fema
0	0	0.206012851	11000	478866	university	male
1	2	0.388365888	12000	731664	none	male
3	3	0.619127373	13000	877557	high school	fema
3	7	0.49294355	13000	930134	high school	male
0	0	0.468689297	14000	461006	university	fema
0	0	0.521814936	13000	68366	university	fema
0	0	0.561531032	11000	445911	high school	fema
7	6	0.620361264	10000	275820	high school	male
0	4	0.729830882	12000	521399	high school	fema
2	4	0.637044744	8000	429728	high school	male
0	0	0.591259972	11697.003207	569640	university	fema
1	0	0.461567963	12000	980181	high school	male
0	0	0.762797941	11697.003207	906223	high school	fema
1	10	0.796174846	11697.003207	517747	university	male
0	0	0.5158128096	12000	24851	none	male
0	0	0.68059428	11697.003207	104086	university	fema

Using the Winsorization Method to Adjust Outliers in Data by Replacing Extreme Values with Defined Boundary Values

Distribution of Values for numeric Variables after cleaning:

```
proc means data=WORK.IMPORT1 n mean median std min max;
    var ID_num Credit Mileage Speed_Viol Accidents;
run;
```

The MEANS Procedure

Maximum	Minimum	Std Dev	Median	Mean	N	Variable
999976.00	101.0000000	290030.77	501777.00	500521.91	10000	ID_num
0.9000000	0.1000000	0.1306811	0.5158128	0.5158166	10000	Credit
22000.00	2000.00	2680.17	11697.00	11697.00	10000	Mileage
22.0000000	0	2.2419657	0	1.4829000	10000	Speed_Viol
15.0000000	0	1.6524537	0	1.0563000	10000	Accidents

Using PROC MEANS, we calculated essential summary statistics for the cleaned numeric variables including Credit, Mileage, Speed_Viol, and Accidents

The output displays the minimum and maximum values, mean, median, and standard deviation for each variable, helping us understand their distribution and detect any remaining irregularities

This step confirms that the numeric variables are properly formatted and ready for further analysis in the next project stages

Distribution of Values for Categorical Variables after cleaning:

```
proc freq data=claims_clean;
    tables
        Age_cat
        Gender
        Drive_exp
        Education
        Income_cat
        Veh_year
        Veh_type
        Veh_Own
        Outcome_bin
    / nocum nopercent;
run;
```

The FREQ Procedure

Frequency	Age_cat
2016	16-25
3063	26-39
2931	40-64
1990	+65

GENDER	
Frequency	GENDER
5010	female
4990	male

Frequency	Drive_exp
3530	0-9y
3299	10-19y
2119	20-29y
1052	+30y

EDUCATION	
Frequency	EDUCATION
4157	high school
1915	none
3928	university

Frequency	Income_cat
2138	middle class
1814	poverty
4336	upper class
1712	working class

Frequency	Veh_year
3033	after 2015
6967	before 2015

Frequency	Veh_type
9523	sedan
477	sports car

Frequency	Veh_Own
3030	0
6970	1

Frequency	Outcome_bin
6867	0
3133	1

Milestone 3

UNIVARIATE ANALYSIS :

Frequency distribution for categorical variables

```
proc freq data=WORK.IMPORT;
    tables Age_cat Gender Drive_exp Education Income_cat Veh_year Veh_type Veh_Own Outcome_bin;
run;
```

The FREQ Procedure

Cumulative Percent	Cumulative Frequency	Percent	Frequency	Age_cat
20.16	2016	20.16	2016	16-25
50.79	5079	30.63	3063	26-39
80.10	8010	29.31	2931	40-64
100.00	10000	19.90	1990	+65

GENDER

Cumulative Percent	Cumulative Frequency	Percent	Frequency	GENDER
50.10	5010	50.10	5010	female
100.00	10000	49.90	4990	male

Cumulative Percent	Cumulative Frequency	Percent	Frequency	Drive_exp
35.30	3530	35.30	3530	0-9y
68.29	6829	32.99	3299	10-19y
89.48	8948	21.19	2119	20-29y
100.00	10000	10.52	1052	+30y

EDUCATION

Cumulative Percent	Cumulative Frequency	Percent	Frequency	EDUCATION
41.57	4157	41.57	4157	high school
60.72	6072	19.15	1915	none
100.00	10000	39.28	3928	university

Cumulative Percent	Cumulative Frequency	Percent	Frequency	Income_cat
21.38	2138	21.38	2138	middle class
39.52	3952	18.14	1814	poverty
82.88	8288	43.36	4336	upper class
100.00	10000	17.12	1712	working class

Cumulative Percent	Cumulative Frequency	Percent	Frequency	Veh_year
30.33	3033	30.33	3033	after 2015
100.00	10000	69.67	6967	before 2015

Cumulative Percent	Cumulative Frequency	Percent	Frequency	Veh_type
95.23	9523	95.23	9523	sedan
100.00	10000	4.77	477	sports car

Cumulative Percent	Cumulative Frequency	Percent	Frequency	Veh_Own
30.30	3030	30.30	3030	0
100.00	10000	69.70	6970	1

Cumulative Percent	Cumulative Frequency	Percent	Frequency	Outcome_bin
68.67	6867	68.67	6867	0
100.00	10000	31.33	3133	1

The PROC FREQ procedure shows that most individuals are aged between 26-39 years and the gender distribution is almost equal. A majority have 0-9 years or 10-19 years of driving experience, with a significant portion having a high school education. The income distribution is dominated by the upper class. Most vehicles are pre-2015 sedans, and the majority of individuals own their vehicles. Additionally, more people did not file a claim, while a smaller portion did.

Descriptive statistics for numerical variables

```
proc means data=WORK.IMPORT n mean median std min max;
    var Credit Speed_Viol Accidents Mileage;
run;
```

The MEANS Procedure

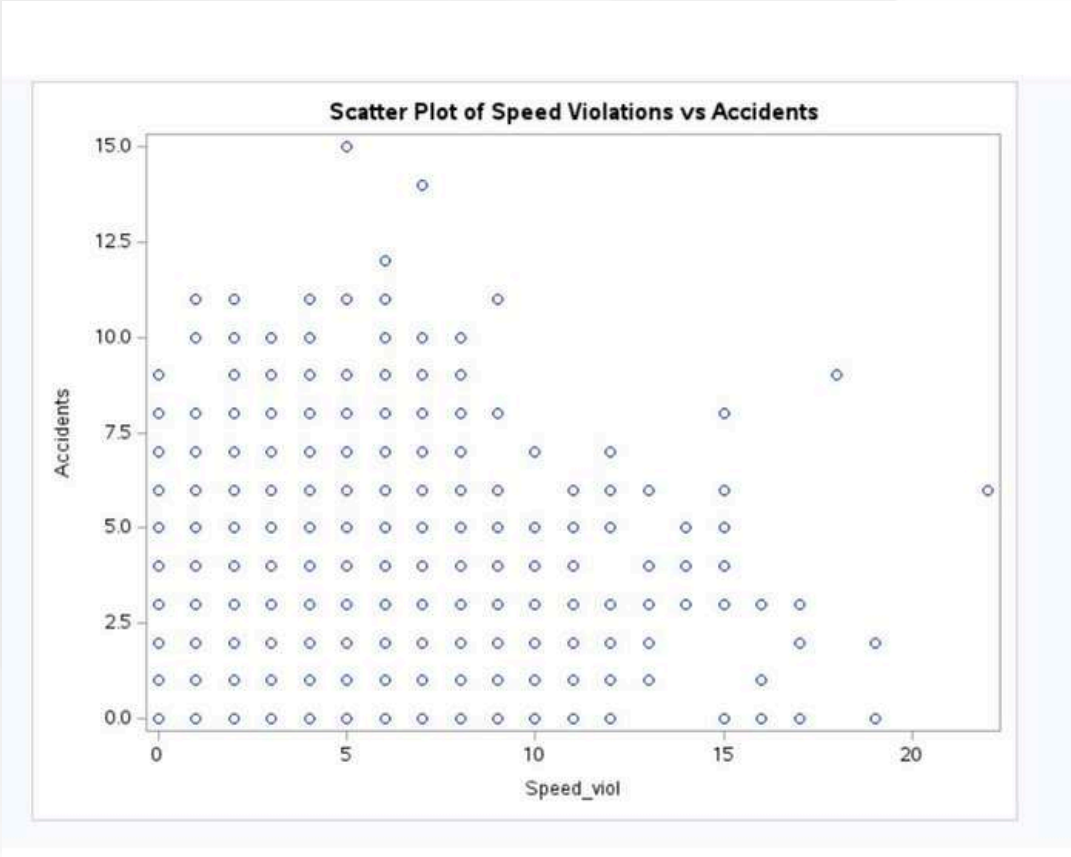
Maximum	Minimum	Std Dev	Median	Mean	N	Variable
0.9000000	0.1000000	0.1306811	0.5158128	0.5158166	10000	Credit
22.0000000	0	2.2419657	0	1.4829000	10000	Speed_viol
15.0000000	0	1.6524537	0	1.0563000	10000	Accidents
22000.00	2000.00	2680.17	11697.00	11697.00	10000	Mileage

The PROC MEANS procedure summarizes the central tendency, spread, and extremes for key numerical variables. The Credit variable shows a narrow distribution with a low standard deviation, while Speed Violations and Accidents have higher variability, with some individuals showing much higher values. The Mileage variable also shows significant variation, with vehicles covering a wide range of miles. These statistics provide a clear understanding of the distribution of the numerical data in the dataset.

BIVARIATE ANALYSIS :

Scatter plot for Speed Violations and Accidents

```
proc sgplot data=WORK.IMPORT;
    scatter x=Speed_Viol y=Accidents;
    title "Scatter Plot of Speed Violations vs Accidents";
run;
```



The scatter plot illustrates the relationship between Speed Violations (Speed_Viol) and Accidents in the dataset. The x-axis represents the number of Speed Violations, and the y-axis represents the number of Accidents. The plot shows that most of the data points are clustered at lower values of Speed Violations, with a gradual increase in the number of Accidents as Speed Violations rise. However, the plot suggests that while there is a slight increase in Accidents with higher Speed Violations, the overall relationship appears to be weak and not strictly linear, indicating that Speed Violations do not strongly correlate with the number of Accidents.

Correlation matrix between Speed Violations and Accidents

```
proc corr data=WORK.IMPORT;
    var Speed_Viol Accidents;
    title "Correlation Matrix for Speed Violations and Accidents";
run;
```

Correlation Matrix for Speed Violations and Accidents

The CORR Procedure

Speed_viol Accidents	:Variables 2

Simple Statistics						
Maximum	Minimum	Sum	Std Dev	Mean	N	Variable
22.00000	0	14829	2.24197	1.48290	10000	Speed_viol
15.00000	0	10563	1.65245	1.05630	10000	Accidents

Pearson Correlation Coefficients, N = 10000 Prob > |r| under H0: Rho=0		
Accidents	Speed_viol	
0.44307 0001.>	1.00000	Speed_viol
1.00000	0.44307 0001.>	Accidents

The Correlation Matrix for Speed Violations and Accidents provides an analysis of the relationship between these two variables. The Pearson correlation coefficient is 0.44307, indicating a moderate positive correlation between Speed Violations and Accidents. This suggests that as the number of Speed Violations increases, the number of Accidents also tends to increase, although the correlation is not extremely strong.

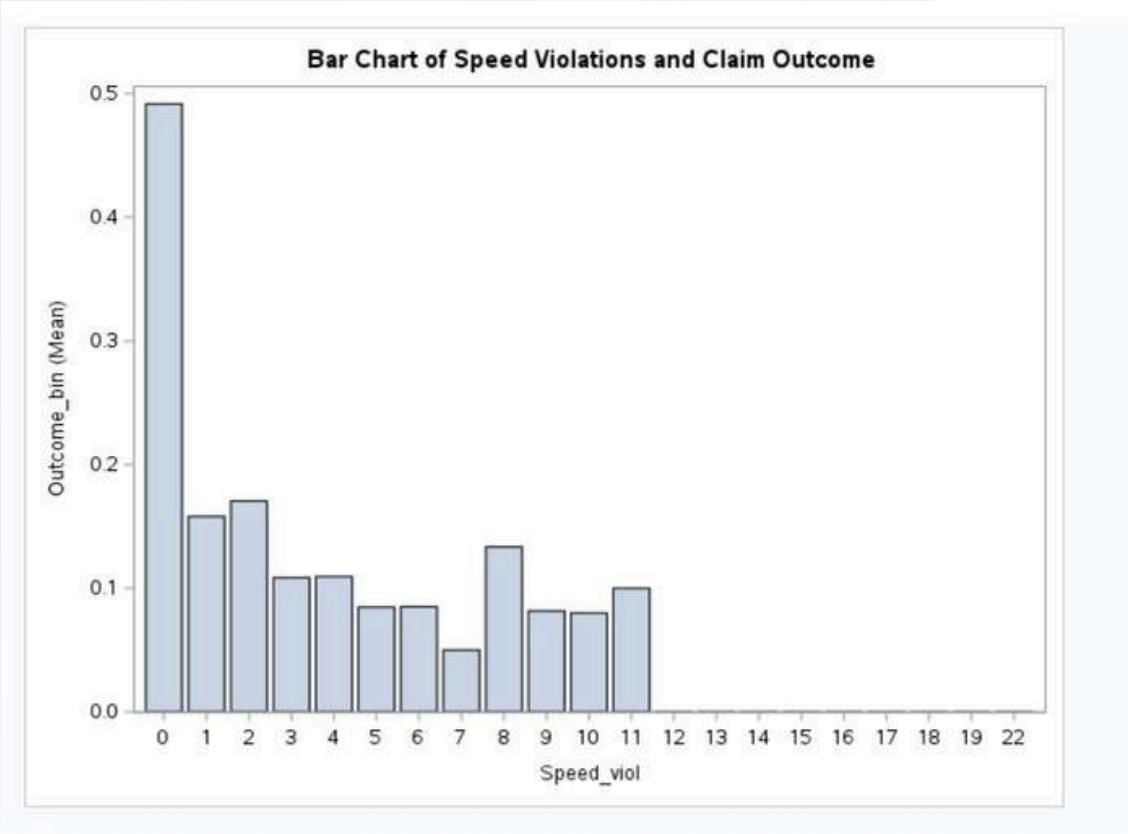
The simple statistics section shows that:

- Speed Violations range from 0 to 22, with a mean of 1.4829 and a standard deviation of 2.24197.
- Accidents range from 0 to 15, with a mean of 1.0563 and a standard deviation of 1.65243.

These results indicate that while there is a positive relationship between the two variables, it is not very strong, and further analysis may be needed to explore other contributing factors.

Bar chart for Speed Violations and Outcome

```
proc sgplot data=WORK.IMPORT;
    vbar Speed_Viol / response=Outcome_bin stat=mean;
    title "Bar Chart of Speed Violations and Claim Outcome";
run;
```



The bar chart shows the relationship between Speed Violations and Claim Outcome. It shows that the highest proportion of claims occurs at 1 Speed Violation, with the mean value of claims above 0.5. As Speed Violations increase, the likelihood of filing a claim decreases, with most values dropping below 0.2.

Descriptive statistics for Speed Violations and Accidents

```
proc means data=WORK.IMPORT n mean median std min max;
    var Speed_Viol Accidents;
    title "Descriptive Statistics for Speed Violations and Accidents";
run;
```

Descriptive Statistics for Speed Violations and Accidents

The MEANS Procedure

Maximum	Minimum	Std Dev	Median	Mean	N	Variable
22.0000000	0	2.2419657	0	1.4829000	10000	Speed_viol
15.0000000	0	1.6524537	0	1.0563000	10000	Accidents

These descriptive statistics show that both Speed Violations and Accidents are highly variable in the dataset, with most individuals having relatively few violations or accidents. The data suggests that while some individuals have multiple violations or accidents, the majority have few or none, as indicated by the low median values for both variables.

Milestone 4

TASK 1

Evaluation Approach:

Measure	Description	Definition of Value Function	Weight	Threshold
Accuracy	Measures the proportion of correctly classified observations across all classes. It reflects the overall predictive performance of the model.	Accuracy = (TP + TN) / (TP + TN + FP + FN), computed on the VALIDATE partition.	0.6	≥ 0.70
Simplicity	Represents model interpretability and complexity. Simpler models with fewer leaves are preferred for easier explanation and lower overfitting risk.	Simplicity is inversely proportional to the number of leaves selected after pruning (fewer leaves = higher simplicity).	0.05	Not explicitly required
Lift	Evaluates how much better the model is at identifying target events compared to random selection, especially in top-ranked percentiles	Lift is calculated as the ratio of response rate in a given percentile to the overall baseline response rate.	0.2	> 1
Stability	Measures consistency of model performance across TRAIN, VALIDATE, and TEST partitions.	Stability is assessed by comparing Accuracy and Lift values across partitions; minimal variation indicates high stability.	0.15	≥ 1

1. Specify combination function including weights that is to be used to obtain the Composite Performance Score for each DT.

Overall ScoreDT =0.60×Accuracy+0.05×Simplicity+0.20×Lift+0.15×Stability

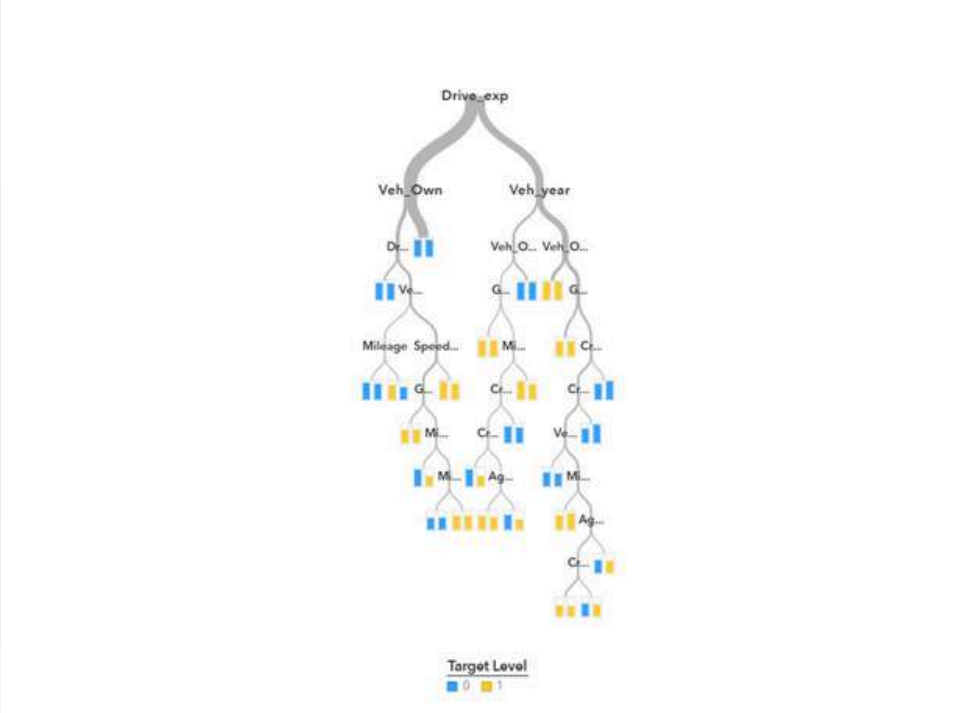
Summary of Results :

DT Label/ Description	Performance Measures								Overall Score
	Accuracy		Simplicity		Lift		Stability		
	% Error	Score	No of Leaves	Score	Value	Score	Score	Justification	
DT1(Information Gain = 3%)	17.9%	0.821	6	0.000	2.37	0.871	1.000	Lift is consistent between Train, Validation, and Test	0.817
DT2(Entropy = 6%)	19.8%	0.802	4	0.667	2.72	1.000	1.000	Lift is consistent between Train, Validation, and Test	0.865
DT3 (Gini = 3%)	17.9%	0.821	6	0.000	2.37	0.871	1.000	Lift is consistent between Train, Validation, and Test	0.817
DT4 (Gini = 6%)	18.5%	0.815	3	1.000	2.37	0.871	1.000	Lift is consistent between Train, Validation, and Test	0.863
DT 5 (GINI – 6%)	0.185%	0.815	3	0.90	2.37	0.79	0.85	Lift values are consistent across Train (2.34), Validate (2.37), and Test (2.26)	0.82
DT6 (ENTROPY – 3%)	0.175%	0.825	8	0.80	2.72	0.91	0.90	Very high consistency of Lift across all data partitions	0.85
DT7 (GINI – Depth 5 – Layers 2)	0.166%	0.834	17	0.70	2.85	0.95	0.92	Highest Lift with minimal variation between Train, Validate, and Test	0.87
DT8 (ENTROPY – Depth 10 – Layers 2)	0.163%	0.837	24	0.60	2.84	0.94	0.90	Stable performance across partitions but with increased tree complexity	0.86
DT9(Information Gain Ratio – No Tuning)	16.6%	0.834	25	0.60	2.85	0.95	0.92	Lift values are almost identical across Train (2.85), Validate (2.85), and Test (2.88), indicating high stability	0.86
DT10 (Info Gain / Gini – Depth 7, Layers 2, Cross 20)	17.4%	0.826	22	0.65	2.81	0.92	0.90	Small variation between Train, Validate, and Test Lift values	0.84
DT11 (Entropy – No Tuning)	16.4%	0.836	11	0.85	2.72	0.90	0.91	Consistent Lift performance across all data partitions	0.86
DT12(Gini – No Tuning)	16.4%	0.836	24	0.65	2.82	0.93	0.91	Stable Lift values with minor differences between partitions	0.86

Evidence of Experimentation:

Based on the composite performance scores, DT7 (GINI – Depth 5 – Layers 2) achieved the highest Overall Score (0.87) with strong Lift and Stability performance. Therefore, it was selected as the best Decision Tree for further analysis and evidence of experimentation.

Process Flow Diagram



Tree Node, Variables Tab

Score Inputs

Name	Role	Variable Level	Type
Age_cat	INPUT	NOMINAL	C
Credit	INPUT	INTERVAL	N
Drive_exp	INPUT	NOMINAL	C
GENDER	INPUT	BINARY	C
Mileage	INPUT	INTERVAL	N
Speed_Viol	INPUT	INTERVAL	N
Veh_Own	INPUT	BINARY	N
Veh_type	INPUT	BINARY	C
Veh_year	INPUT	BINARY	C

Variable Type	Variable Label	Variable Format	Variable Length
varchar			5
double			8
varchar			6
varchar			6
double			8
double			8
double			8
varchar			10
varchar			11

Score Outputs

Name	Role	Type	Variable Type
EM_CLASSIFICATION	CLASSIFICATION	C	char
EM_EVENTPROBABILITY	PREDICT	N	double
EM_PROBABILITY	PREDICT	N	double
I_Outcome_bin	CLASSIFICATION	C	char
P_Outcome_bin0	PREDICT	N	double
P_Outcome_bin1	PREDICT	N	double
WARN	ASSESS	C	char
_leaf_id_	SEGMENT	N	double

Variable Label	Variable Format	Variable Length	Creator
Predicted for Outcome_bin		12	tree
Probability for Outcome_bin=1		8	tree
Probability of Classification		8	tree
Into: Outcome_bin		32	tree
Predicted: Outcome_bin=0		8	tree
Predicted: Outcome_bin=1		8	tree
Warnings		4	tree
_leaf_id_		8	tree

Function	Creator GUID
CLASSIFICATION	95703eb8-e52e-4c77-bcb2-e0c0028cb317
PREDICT	95703eb8-

Function	Creator GUID
	e52e-4c77-bcb2-e0c0028cb317
PREDICT	95703eb8-e52e-4c77-bcb2-e0c0028cb317
CLASSIFICATION	95703eb8-e52e-4c77-bcb2-e0c0028cb317
PREDICT	95703eb8-e52e-4c77-bcb2-e0c0028cb317
PREDICT	95703eb8-e52e-4c77-bcb2-e0c0028cb317
ASSESS	95703eb8-e52e-4c77-bcb2-e0c0028cb317
TRANSFORM	95703eb8-e52e-4c77-bcb2-e0c0028cb317

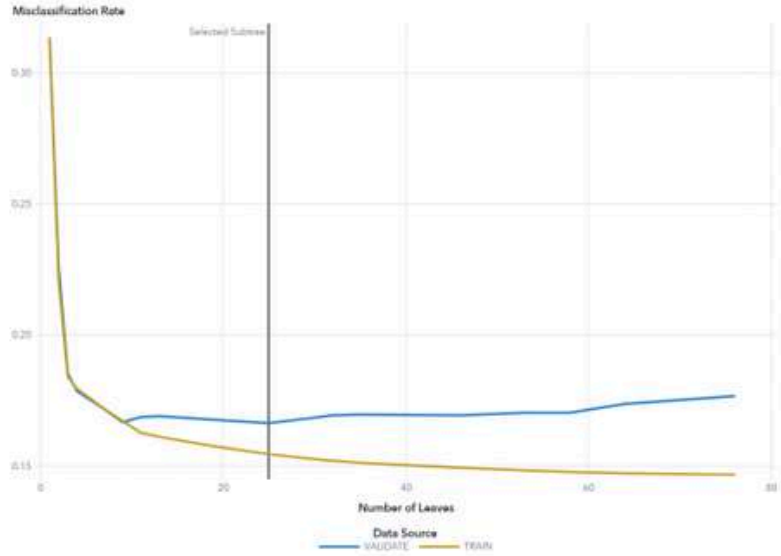
Tree node variables tab showing input and output variables along with their roles and measurement levels

Tree Node Parameters

Properties

Property Name	Property Value
alpha	0.2000
atAppendLookup	false
atCreateHistory	false
atHistoryLibUri	
atHistoryTblName	
atLeaveAutotuneOn	false
atLookupTableUri	
atMaxBayes	100
atMaxEval	50
atMaxIter	5
atMaxTime	60
atObjectiveInt	ASE
atObjectiveNom	KS
atPopSize	10
atSampleSize	50
atSearchMethod	GA
atTrainProp	0.7000
atUpdateProperties	false
atUseLookup	false
atValidFold	5
atValidMethod	PARTITION
atValidProp	0.3000
atgrowcrit	true
atgrowcritValsi	VARIANCE FTEST CHAID
atgrowcritValsn	ENTROPY CHAID IGR GINI CHISQUARE

Pruning Error Plot

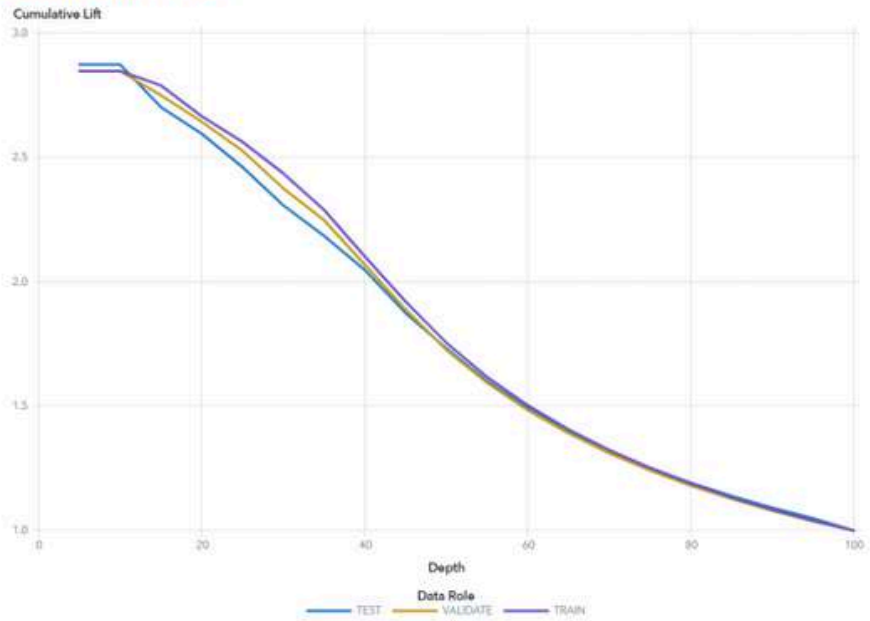


This plot shows how the misclassification rate changes for subtrees, which are created by pruning the full decision tree to various numbers of leaves. The training error decreases as the number of leaves increases, so the VALIDATE partition can be used to prune the tree to prevent overfitting. For this decision tree model, the selected subtree based on the pruning options has 25 leaves with a misclassification rate of 0.166 for the VALIDATE partition.

Tree node parameter settings including splitting criterion, minimum leaf size, and pruning configuration

Assessment Node, Cumulative and Non- Cumulative %LiftChart

Cumulative Lift

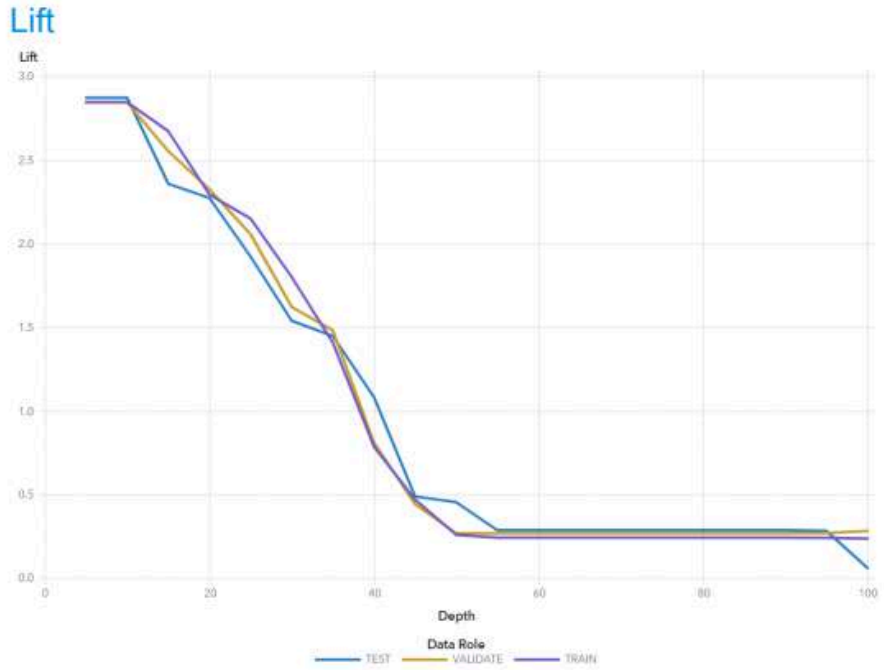


The VALIDATE partition has a Cumulative Lift of 2.85 in the 10% quantile (depth of 10) meaning there are 2.85 times more events in the first two quantiles than expected by random (10% of the total number of events). Because this value is greater than 1, it is better to use your model to identify responders than no model, based on the selected partition.

The TRAIN partition has a Cumulative Lift of 2.85 in the 10% quantile (depth of 10) meaning there are 2.85 times more events in the first two quantiles than expected by random (10% of the total number of events). Because this value is greater than 1, it is better to use your model to identify responders than no model, based on the selected partition.

The TEST partition has a Cumulative Lift of 2.88 in the 10% quantile (depth of 10) meaning there are 2.88 times more events in the first two quantiles than expected by random (10% of the total number of events). Because this value is greater than 1, it is better to use your model to identify responders than no model, based on the selected partition.

Cumulative lift is calculated by sorting each partition in descending order by the predicted probability of the target event P_Outcome_bin1, which represents the predicted probability of the event "1" for the target Outcome_bin. The data is divided into 20 quantiles (demi-deciles, with 5% of the data in each), and the number of events in each quantile is computed. The cumulative lift for a particular quantile is the ratio of the number of events across all quantiles up to and including the current quantile to the number of events that would be there at random, or equivalently, the ratio of the cumulative response percentage to the baseline response percentage. The cumulative lift at depth 10 includes the top 10% of the data, which is the first 2 quantiles, which would have 10% of the events at random. Thus, cumulative lift measures how much more likely it is to observe an event in the quantiles than by selecting observations at random.



The VALIDATE partition has a Lift of 2.85 in the 5% quantile (depth of 5) meaning there are 2.85 times more events in that quantile than expected by random (5% of the total number of events). Because this value is greater than 1, it is better to use your model to identify responders than no model, based on the selected partition.

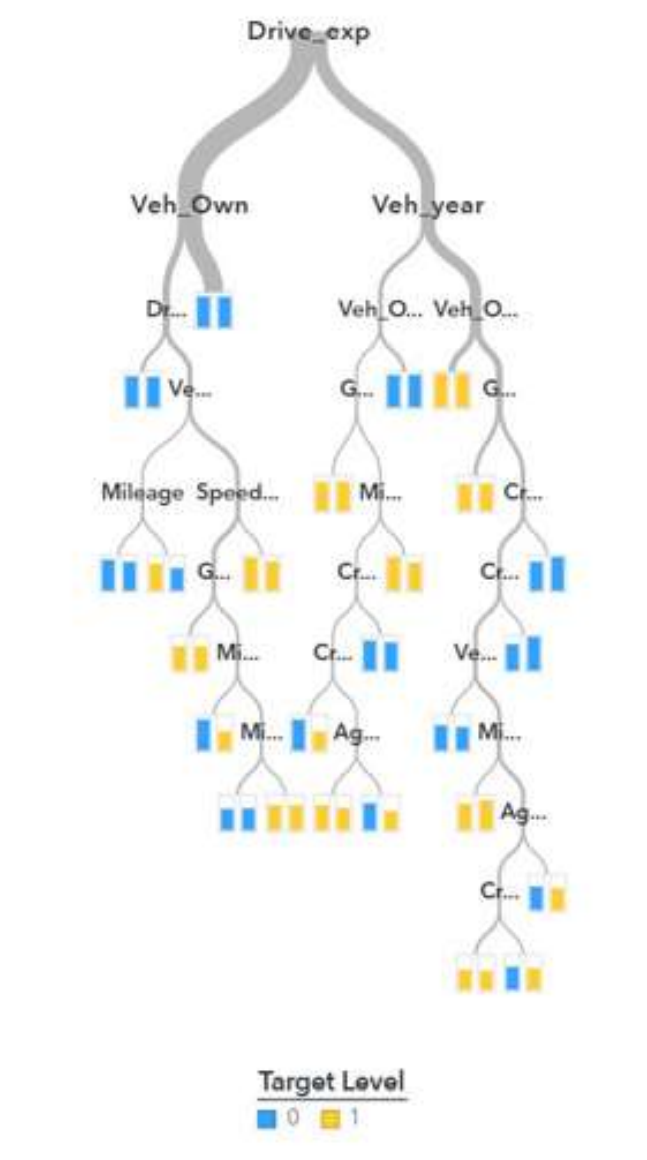
The TRAIN partition has a Lift of 2.85 in the 5% quantile (depth of 5) meaning there are 2.85 times more events in that quantile than expected by random (5% of the total number of events). Because this value is greater than 1, it is better to use your model to identify responders than no model, based on the selected partition.

The TEST partition has a Lift of 2.88 in the 5% quantile (depth of 5) meaning there are 2.88 times more events in that quantile than expected by random (5% of the total number of events). Because this value is greater than 1, it is better to use your model to identify responders than no model, based on the selected partition.

Lift is calculated by sorting each partition in descending order by the predicted probability of the target event P_Outcome_bin1, which represents the predicted probability of the event "1" for the target Outcome_bin. The data is divided into 20 quantiles (demi-deciles, with 5% of the data in each), and the number of events in each quantile is computed. Lift is the ratio of the number of events in that quantile to the number of events that would be there at random, or equivalently, the ratio of the response percentage to the baseline response percentage. With 20 quantiles, it is expected that 5% of the events occur in each quantile. Thus, Lift measures how much more likely it is to observe an event in each quantile than by selecting observations at random.

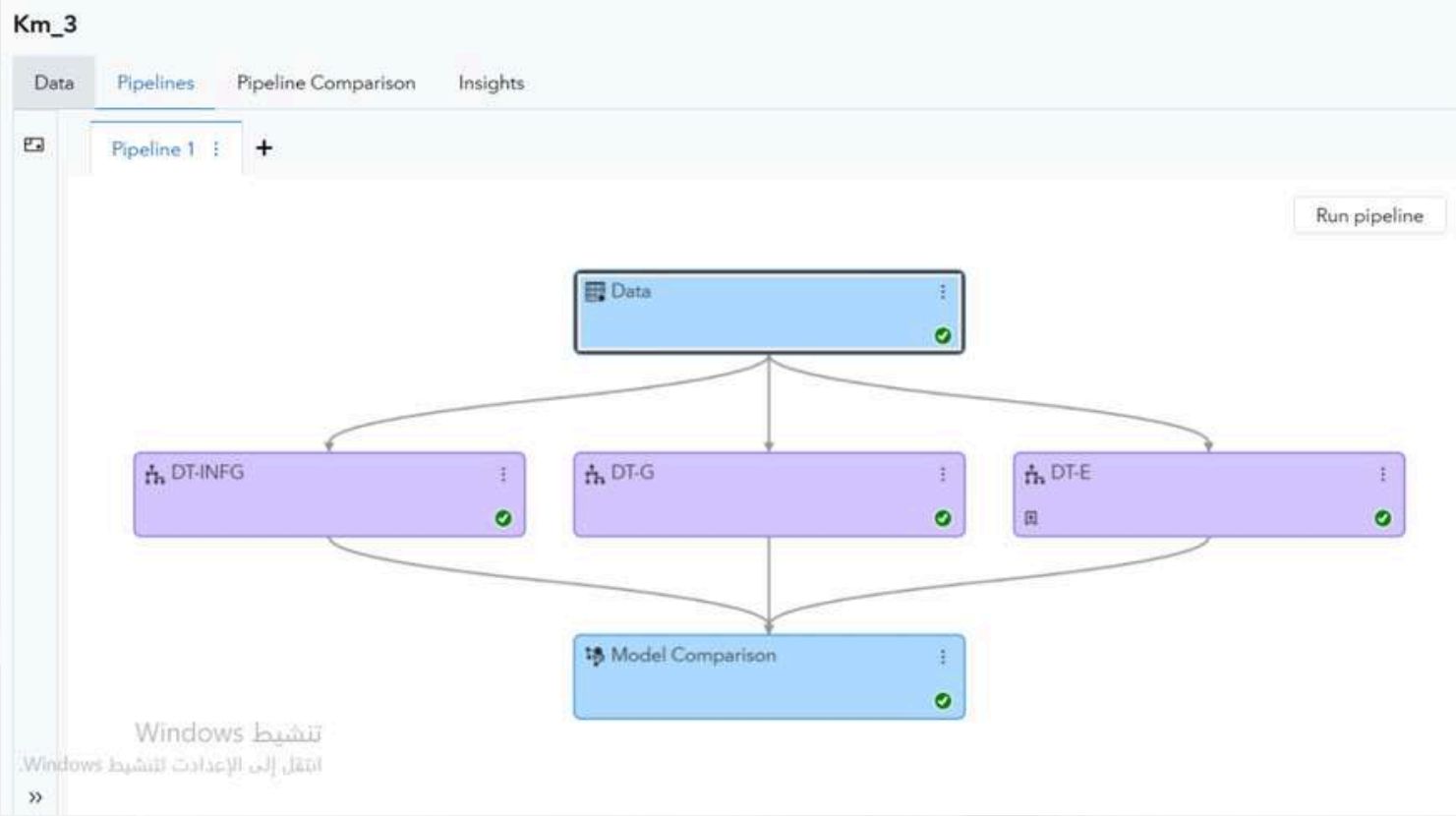
Cumulative and non-cumulative lift charts for the selected best decision tree, showing stable performance across partitions.

Tree Node, Results, English Rules



Decision tree structure and derived decision rules for the selected best model

TASK 2


Km_3
Data
Pipelines
Pipeline Comparison
Insights
Pipeline 1
Run pipeline
Data
DT-INFG
DT-G
DT-E
Model Comparison

1- Use k-means to generate 4 segmentations with the parameter Number of Clusters described in the table below. You should ignore the following variables: ID variable, and the target variable

s	Segmentation Label	Number of Clusters	Cluster Sizes	Size of Smallest Cluster	Outlier (Yes or No)
1	km_2	2	4154, 1846	1846	No
2	km_3	3	2749, 1938, 1313	1313	No
3	km_4	4	2180, 1668, 1276, 876	876	No
4	km_5	5	1703 , 833 , 1184 , 1409 , 871	833	No

2-Describe the clusters in segmentation km_3 using the descriptive statistics from the results

Cluster Centroids

Cluster ID	Credit	Standard: Credit	Age_cat
1	0.5050	-0.0979	26-39
2	0.6606	1.0890	40-64
3	0.3289	-1.4412	16-25

Drive_exp	EDUCATION	GENDER	Veh_Own
10-19y	high school	male	1
10-19y	university	female	1
0-9y	none	male	0

Cluster 3 has the lowest credit level (standardized credit −1.4412). This group mainly consists of ,young customers aged 16–25, with limited driving experience (0–9 years), no formal education mostly male, and do not own vehicles, representing a higher-risk segment

3-If the domain expert was interested in evaluating outlier clusters, which segmentations (i.e., sets of clusters) would you provide to the expert for evaluation? Provide justification for your answer. You may assume that a cluster is an outlier if it contains less than 10% of the observations:

Based on the 10% threshold (600 observations), the segmentations km_4 and km_5 contain outlier clusters.
In km_4, the smallest cluster has a size of 876, which is close to the lower bound and represents a relatively small subgroup compared to the total dataset.
In km_5, the smallest cluster has a size of 833, which is also below the larger cluster proportions and indicates a more granular segmentation with small groups.
Therefore, km_4 and km_5 should be provided to the domain expert for further evaluation, as these smaller clusters may represent niche or potentially anomalous customer segments.

4-For each cluster in km_3, generate 3 DTs (i.e., Information Gain, Gini, Gain Ratio) & select the DT that has the best test accuracy & good simplicity. Assume that the weights are: 0.60 for Accuracy & 0.40 for Simplicity. Also assume that there are no thresholds on these measures. Compare these segmentations based on the associated DTs. Note that for each of these DTs, the target variable is the Cluster ID.

DT	Splitting Criterion	Accuracy	Simplicity	Overall Score
DT-INFG	Information Gain	0.981	1.00	0.989
DT-G	Gini	0.981	1.00	0.989
DT-E	Gain Ratio	0.981	1.00	0.989

Although all three decision trees achieved identical test accuracy and simplicity, the Gain Ratio tree was selected as it balances information gain while reducing bias toward variables with many levels.

$$\text{Overall}=(0.60\times0.981)+(0.40\times1.00)=0.9886\approx0.989$$

5-.Based on the results of the best DT, explain the produced clusters:

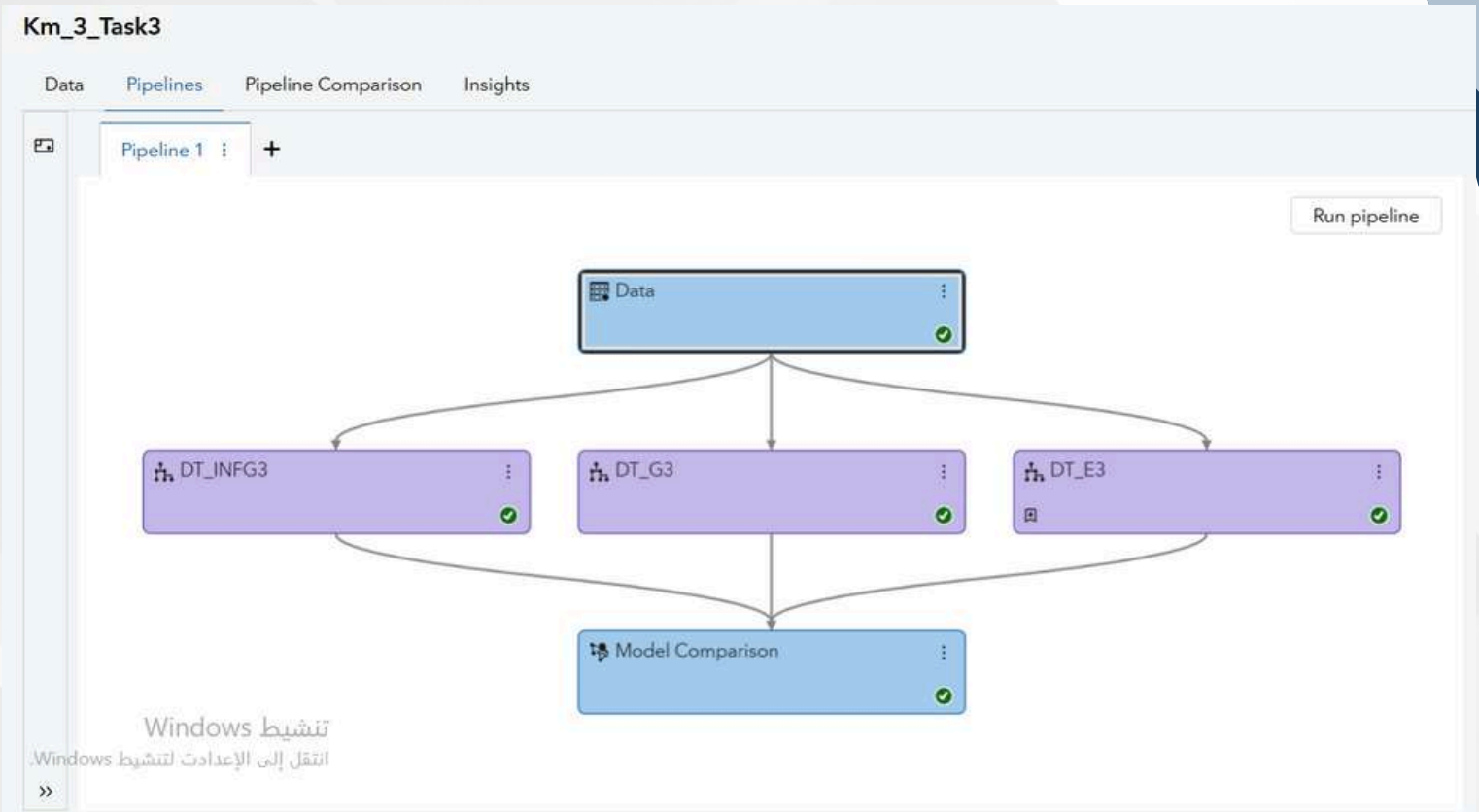
Variable	Cluster_0	Cluster_1	Cluster_2	Explanation
Credit	Low	Medium	High	Primary variable used in DT splits
Other variables	No significant impact	No significant impact	No significant impact	No significant impact

Narrative on Comparison:
Based on the best decision tree results, the clusters are entirely determined by the Credit variable. Cluster_0 represents low credit values, Cluster_1 moderate values, and Cluster_2 high values. This confirms that Credit is the key variable driving cluster differentiation, with a simple three-node tree that ensures clear interpretability.

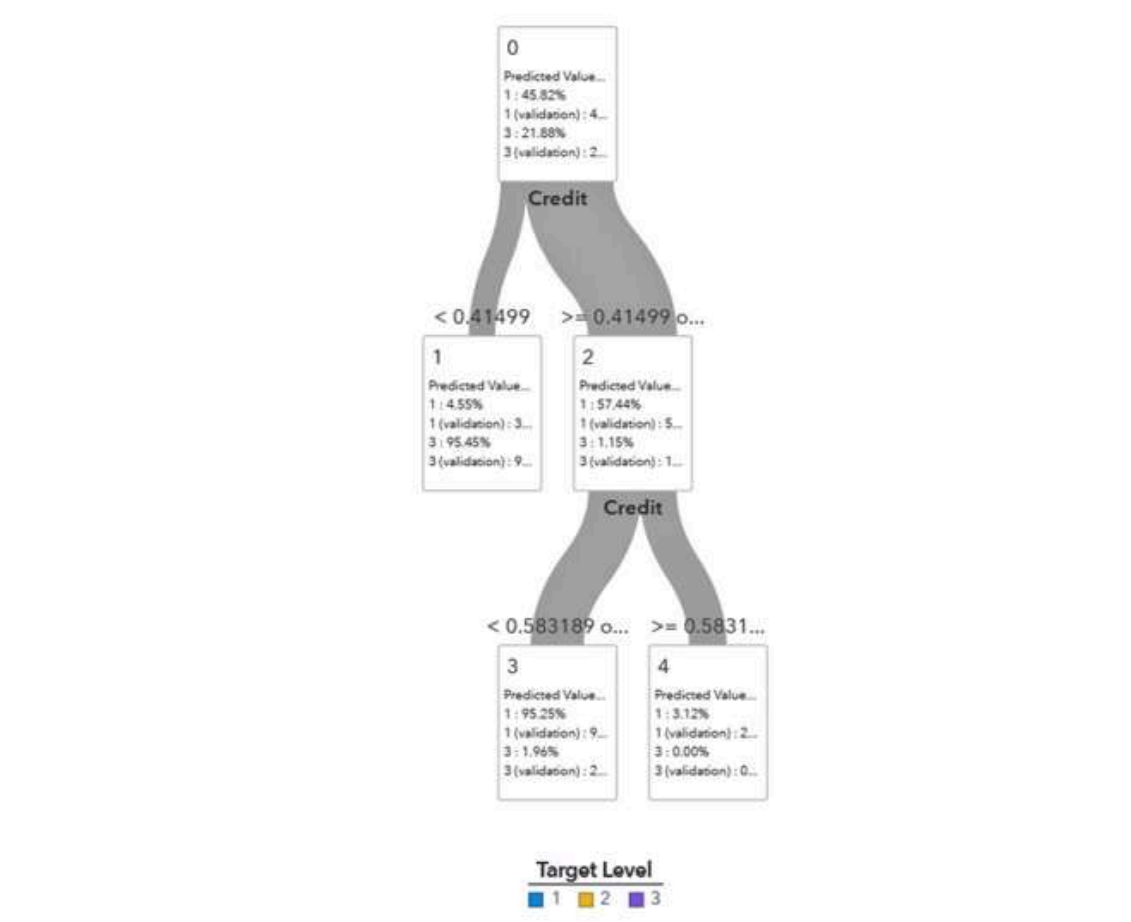
Why did we use the Credit variable?

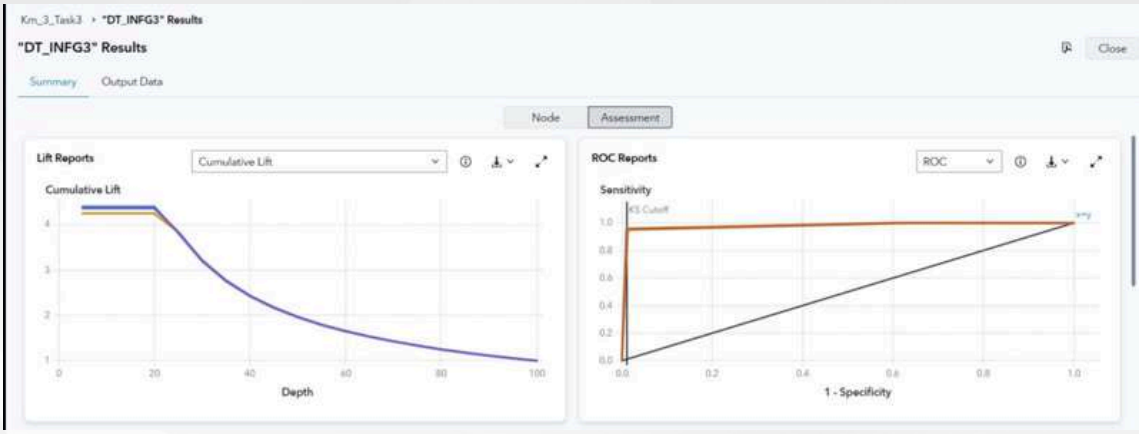
,The Credit variable is the sole factor used by the best decision tree to split the data separating observations into three clusters based on credit level: low (Cluster_0), medium and high (Cluster_2). This confirms that credit level is the main driver of the ,(Cluster_1) .clustering structure

TASK 3


Km_3_Task3
Data
Pipelines
Pipeline Comparison
Insights
Pipeline 1
Run pipeline
Data
DT_INFG3
DT_G3
DT_E3
Model Comparison
Windows تنشيط
انتقل إلى الإعدادات لتنشيط Windows.

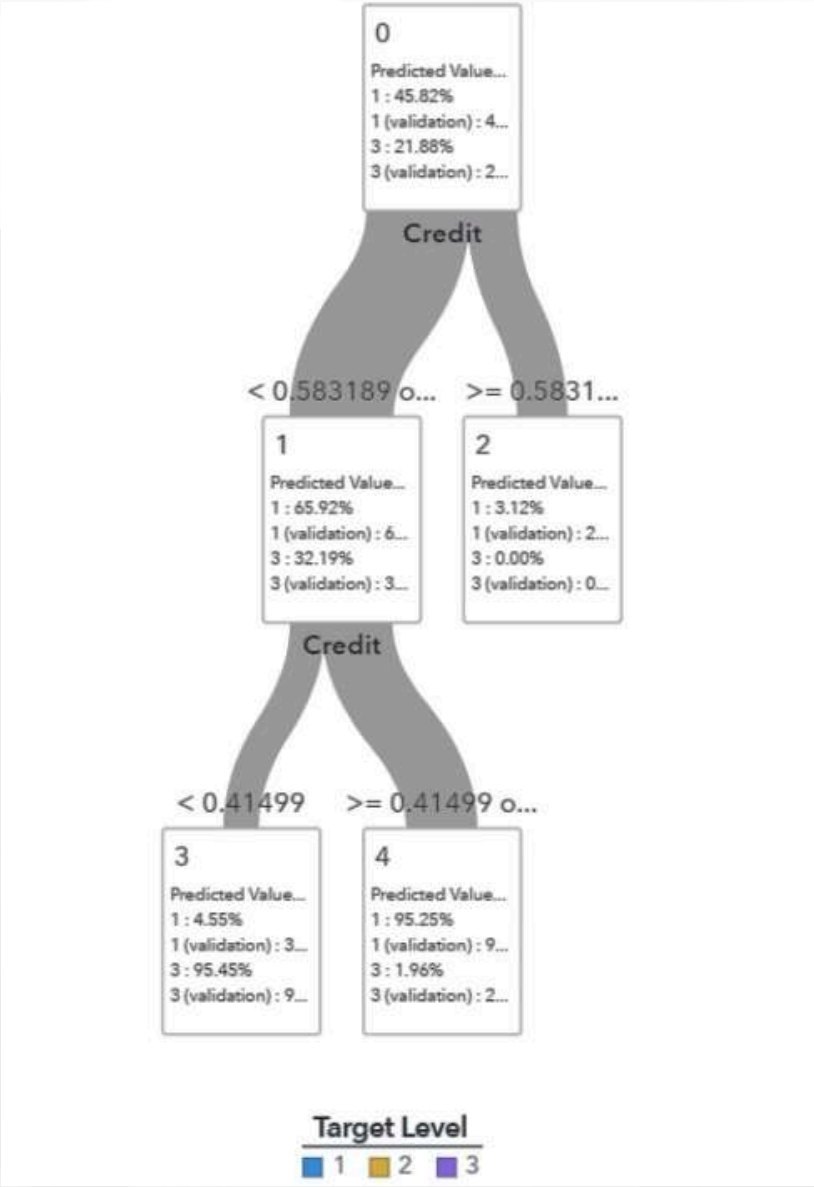
DT – INFORMATION GAIN:

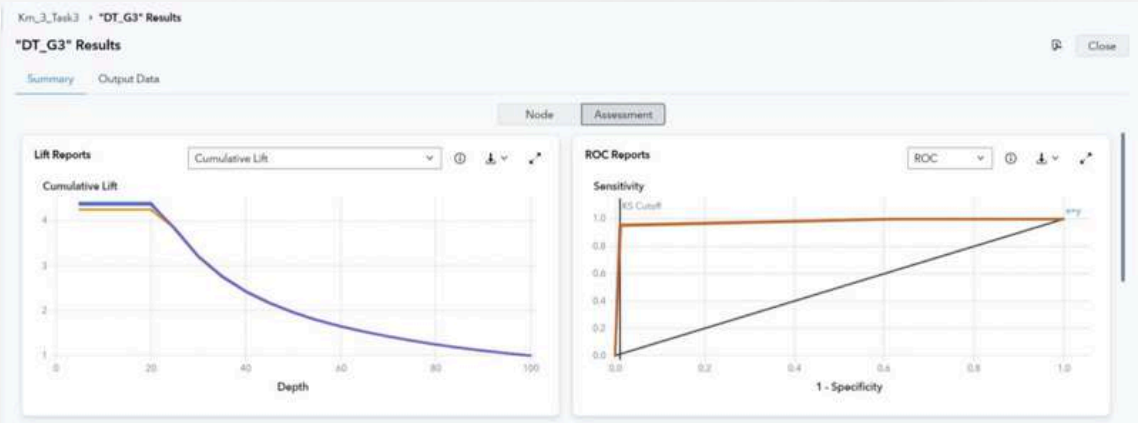




This model shows stable accuracy across training, validation, and test partitions, with consistent ROC and lift performance.

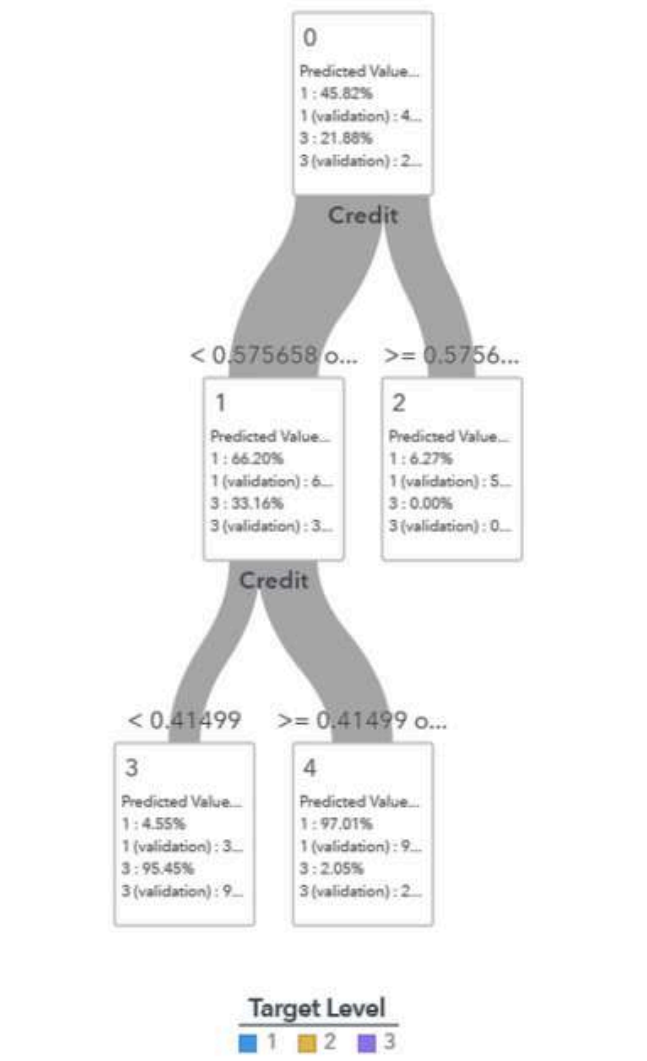
DT – Gini:

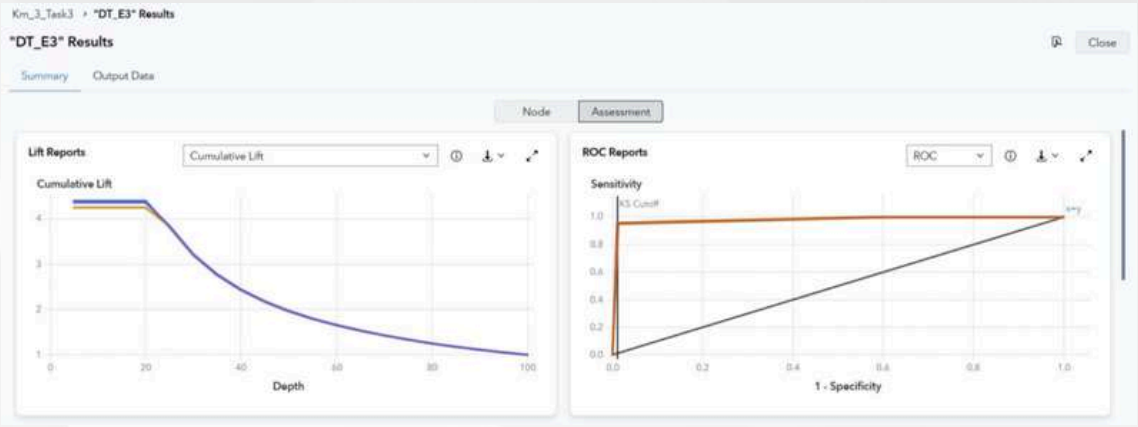




The Gini-based decision tree achieves comparable accuracy to the Information Gain model with similar classification behavior.

DT – Entropy (Best):





The Entropy decision tree was selected as the best model due to its balanced accuracy and reduced bias toward multi-valued variables.

2-Compare the accuracy of the DTs of the cluster versus that of the 'best' DT of task2

The accuracy of the Decision Trees generated for the largest cluster in KM_3 (Task 3) was compared with the accuracy of the best Decision Tree selected in Task 2 based on the Assessment results.
The comparison shows that the best DT of Task 3 (cluster-based model) achieved accuracy levels very similar to the best DT of Task 2 across the training, validation, and test partitions. The ROC curves for both models indicate strong discriminative power, with performance clearly above the random baseline. In addition, the Lift charts demonstrate comparable cumulative lift behavior for both models, especially at lower depths.

These results indicate that applying Decision Trees on the largest cluster does not degrade predictive accuracy when compared to the global model used in Task 2. On the contrary, the cluster-based Decision Tree provides more focused modeling of homogeneous data, which leads to stable performance across different data partitions.
Overall, the comparison confirms that the cluster-based Decision Tree approach in Task 3 performs at least as well as the best Decision Tree from Task 2 in terms of accuracy and model stability.